

IT3050 - Employability Skills Development



Mid-Semester Assignment

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1. Evidence-Based Self-Reflection & Lecture Impact Analysis

A major event that showed my employability preparedness happened when I was doing my Year 2 group project on Event Planning Systems. The system had several modules which included event management, booking and user interaction modules. I was to create the Feedback module, Admin Reply, where users were to leave feedback and the administrators to reply to the feedback in the system.

The combination of all modules in the integration stage was a significant challenge that we encountered. Although my features were functioning as expected on their own, they seemed to have problems when they were combined with other parts. In particular, the feedback data was not properly manifested on the admin dashboard because of the inappropriate structures of API responses and irregularities in processing the data on the frontend and the backend.

This resulted in confusion in the team and slowed down the process because the other members would require precise feedback data. I thought that the problem was with other modules initially but after debugging I noticed that some of my API responses were not in an integration format.

I also made it my task to fix the problem, collaborating with the frontend developer. We harmonized the API formats, fixed the response formats and did repeated testing. Luckily, we were able to resolve the problem and finish the project on schedule.

This experience showed that I am able to work under pressure to troubleshoot technical problems and be responsible. However, it also showed that I need to work on the aspect of early integration testing, proactive communication, and system level thinking. It also stressed the point that employability is not merely about individual talents in code, but also to ensure that all parts of a system have been linked up.

Self-Rating (1–5) with Evidence:

- **Accountability – 4/5:**

I took full responsibility about identifying and fixing the integration in my module.

Risk: May take on too much pressure without work effectively.

- **Communication – 3/5:**

I communicated with the frontend developer during debugging, but only after the issue became serious.

Risk: Delayed communication can heading many issues in a real work environment.

- **Teamwork – 4/5:**

Collaborated effectively with team members during integration and testing.

Risk: Sometimes focus more on my own tasks than overall team coordination.

- **Adaptability – 4/5:**

Adapted easily to integration issues and how to coordinate systems.

Risk: Like to respond to issues rather than predict them in advance.

- **Professional Discipline – 3/5:**

Completed tasks on time but lacked early validation of integration compatibility.

Risk: Absence of early testing paths may influence the quality of products in a professional environment.

Professional Responsibility vs Task Completion is one such lecture that impressed me a great deal. The point was that it is not sufficient to finish works but it is necessary that professionals should take care about the quality, reliability, and the usability of their work.

This concept was a challenge to my past way of thinking since I was primarily worried about fulfilling my given module. This lecture made me understand that although my code is effective as an individual, it needs to be in harmony with the rest of the system. This transformed my mindset in the way I laid emphasis on quality assurance, collaboration and responsibility in software development.

2. Industry Alignment & Professional Judgment

In this analysis, I have taken into consideration a position of a Junior Software Engineer in a company dealing with software development.

Three required skills:

1. Programming skills (Java/JavaScript)

I have work experience in academic projects like the Event Planning System and MERN based applications. I must work on writing clean, maintainable, and scalable code.

2. Database management (SQL)

I have also executed my skills to design databases and write queries. My integration problem, however, demonstrated that I have to work on data consistency and validation across systems.

3. Problem-solving and debugging

I trust my ability to debug problems, and most importantly when it is critical. Nevertheless, I occasionally use reactive techniques rather than methodical methods of debugging.

Ethical Scenario Response:

In case I found a serious bug towards a deadline and my Team Lead recommended that we should look back it to prevent delays, I would bring up the question and make suggestions on how to rectify it before launch. The release would be not responsible in case the bug has any impact on the functionality of a system or user experience.

I consider this choice as an expression of my professional values of integrity, accountability, and professionalism. Providing a defective product with awareness can hurt user confidence and reputation of the company.

I think that I am gaining the courage to be a professional in stressful situations. Though this might be difficult to say to a Team Lead, I would do this in a respectful way, explain the risks involved and suggest some possible ways to rectify it such as making it a priority to fix it or reporting the issue to the stakeholders.

3. My Immediate Upgrade Plan (Within 6 months)

In order to become more employable ready I will undertake the following:

1. Create 2 full-stack (MERN or Spring Boot) projects fully integrated and documented.
 - Timeline: 4 months
 - Outcome: Improve system integration skills and coding standards
2. Design database and optimize SQL by solving at least 100 real-life queries and scenarios.
 - Timeline: 3 months
 - Result: Develop data processing and integrity.
3. Complete at least 3 mock interviews and practice presentations.
 - Timeline: 6 months
 - Result: Become a better communicator, confident, and professional.

These measures are very specific, measurable and directly target my present areas of weakness in integration, communication, and professional discipline.

Conclusion

According to my experiences gained during the last two years, I think that I am moderately employable but not industry-ready. I possess good technical backgrounds and problem solving skills, however, I still require to work on active communication, early testing, and system level thinking. Through my upgrade plan, I believe I will be able to achieve a high level of employability preparedness and a responsible and professional software engineer.